PRINCE MODI

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EDUCATION

University of California, San Diego Sep 2025 – Jun 2027
MS Computer Science | GPA: 3.88/4.00
- Relevant Coursework: Distributed Systems, Deep Learning Systems, LLM System Optimization

Ganpat University, India Aug 2018 – Jul 2022
B. Tech. Computer Engineering | GPA: 3.96/4.00
- Relevant Coursework: Operating Systems, Cloud Computing **| Academic Scholarship |** 2nd rank out of 60+ students

WORK EXPERIENCE

Picasso Lab, UC San Diego Jul 2026 – Present
Graduate Student Researcher, LLM Inference Systems
- Profiling the attention operator in transformer **LLM inference** across prefill (prompt/KV-cache lengths) and decode (batch size/KV-cache lengths), fitting analytical models to measured compute and communication latencies
- Building a trace-driven simulator over multi-turn **LLM** conversation traces to compare parallelism strategies (CP/TP/DP + EP) for a disaggregated prefill/decode serving deployment

Indian Institute of Science (IISc), Bangalore Jan 2023–Apr 2024 (Full-time); Apr 2024–Nov 2024 (Voluntary)
Project Associate, Distributed Systems
Flotilla: Distributed Federated Learning Framework
- Architected a distributed **federated-learning** framework (Python/AsyncIO) for **privacy-preserving model training** across **1,000+ workers**, implementing synchronous and asynchronous aggregation (FedAvg, FedAsync, tiered) via a pluggable event-driven state machine
- Designed a stateful parameter-server architecture with stateless workers, externalizing all orchestration state to **Redis** for real-time durability, cutting per-round overhead to just **1.7%** at 1,000+ nodes vs. **54.9%** for a leading open-source baseline
- Built **gRPC + MQTT** worker discovery and heartbeat-driven liveness detection, and implemented server-side fault tolerance via incremental checkpointing and hot failover, recovering a live session on a standby server within **820ms** with rollback limited to a single round
- Validated **on-device training** across heterogeneous edge accelerators (Nvidia Jetson GPUs, ARM devices) for CNN, LSTM, and transformer (ALBERT) models in **PyTorch/HuggingFace**, tolerating **~40% worker dropout** with negligible accuracy loss

Sterlite Technologies Limited, Ahmedabad Jan 2022 – Aug 2022
Software Engineering Intern
- Refactored Docker images on Linux CI/CD pipelines, reducing image size by **35%** and accelerating build times by **50%**
- Integrated a Liquibase module to track MongoDB schema changes, improving version control reliability across environments

PUBLICATION

Prince Modi, Varad Kottawar, Ayush Govind. **Recovering Internal Command Interfaces from Applications for Direct Agent Invocation.** *SOSP 2026 Poster Session (to appear)*

Roopkatha B., **Prince Modi**, et al. **Flotilla: A Scalable, Modular and Resilient Federated Learning Framework for Heterogeneous Resources.** *Journal of Parallel and Distributed Computing (JPDC 2025)*

PROJECTS

Distributed Container Snapshotting | *Podman, CRIU* June 2026
- Designed a distributed **snapshotting system** for stateful containerized applications, implementing the **Chandy-Lamport algorithm** to capture globally consistent cuts across nodes communicating over UDP
- Built a **reliable-UDP sidecar proxy** using sequence numbering, buffering, and acknowledgment retransmission to guarantee in-order, exactly-once delivery without modifying application code

FlashAttention Kernel | *Triton, Python* Feb 2026
- Implementing **FlashAttention** and RMSNorm kernels in **Triton**, applying **tiled softmax** and **online normalization** to minimize HBM memory accesses and improve throughput for long-context LLM inference

TECHNICAL SKILLS

- Machine Learning: PyTorch, HuggingFace, SGLang, Triton, NCCL
- Languages: Python (proficient), C, Java, Bash/Shell
- Distributed Systems: gRPC, MQTT, Redis, Fault-Tolerant Orchestration, Checkpoint/Restore
- Infrastructure: Linux, Docker/Podman, CRIU, PostgreSQL, Git